NEARLY REGULAR DECOMPOSITIONS OF GRAPHS

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Abstract. A conjecture by A. Hoffman-Ostenhof suggests that any connected cubic graph G contains a spanning tree T for which every component of $G - E(T)$ is a K_1, a K_2, or a cycle. We show that any cubic graph G contains a spanning forest F for which every component of $G - E(F)$ is a K_2 or a cycle, and that any connected graph $G \neq K_1$ with maximal degree at most 3 contains a spanning forest F without isolated vertices for which every component of $G - E(F)$ is a K_1, a K_2 or a cycle. We also prove a related statement about path-factorizations of graphs with maximal degree 3.

1. Introduction

Let $G = (V, E)$ be a simple undirected graph. If $|V| = 1$, then G is *trivial*. An edge incident to a vertex of degree one is a *pendant* edge. In general a *bridge* is an edge e such that $G - e$ has more components than G. For $k \in \mathbb{N}$ a *k-vertex* is a vertex of degree k. The subgraph $G[X]$ *induced* by $X \subseteq V$ in G has vertex set X and as its edge set the set of edges of G both ends of which belong to X. Let $A \subseteq E$. Then similarly, the subgraph $G[A]$ *induced* by A has edge set A, and its vertex set is the set of those vertices of G that are ends of at least one edge belonging to A. A *spanning subgraph* of G is a subgraph $(V, E \setminus A)$ for some subset A of E. This subgraph is also denoted by $G - A$.

Let $a, b \in \mathbb{N}$ with $a \leq b$. If $d(v) \in \{a, b\}$ holds for every $v \in V$, then G is an *$[a, b]$-graph.* If every component of G is either a-regular or b-regular, then G is an *$\{a, b\}$-graph.* A subgraph $F \subseteq G$ is an *$[a, b]$-factor,* respectively an *$\{a, b\}$-factor* of G if F is an $[a, b]$-graph, respectively $\{a, b\}$-graph, with vertex set V. For $a = b = r$ we omit the brackets and write r-factor.

Observe that a $\{1, 2\}$-factor in a graph is a spanning subgraph consisting of the union of disjoint edges and cycles, while a $[1, 2]$-factor is a spanning subgraph consisting of the union of disjoint (meaning sharing no vertices) paths and cycles. A 3-regular graph is also called *cubic*. A *matching* in G is a set M of pairwise disjoint edges, it is *perfect* if $G[M]$ is spanning.

The following conjecture is due to A. Hoffman-Ostenhof (see [9]).

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Conjecture A. *Every connected cubic graph G contains a spanning tree T for which each component of $G - E(T)$ is an isolated vertex, an isolated edge, or a cycle.*

This is clearly equivalent to the statement that every connected cubic graph can be decomposed into a spanning tree and a $\{1, 2\}$-factor of a subgraph.

In [6], Mazzuoccolo conjectured several statements related to the Berge-Fulkerson Conjecture, which posits that any cubic graph has six perfect matchings such that each edge is in exactly two. In particular, he conjectured the following which would be a consequence of the Berge-Fulkerson Conjecture.

Conjecture B. *For every bridgeless cubic graph, there are two perfect matchings such that the complement of their union is a bipartite graph.*

This is equivalent to the statement that every bridgeless cubic graph can be decomposed into a spanning bipartite graph and a bipartite $\{1, 2\}$-factor.

Motivated by these and several other problems raised in [6], we study various decomposition properties of graphs of restricted degrees. To give the reader some context to our results, we now observe how their restrictions to cubic graphs are related to Conjectures A and B.

Our Observation 1 implies that every cubic graph can be decomposed into a spanning bipartite graph and a bipartite $[1, 2]$-factor, giving a weakening of Conjecture B. Observation 5 strengthens this by saying that both subgraphs in the decomposition can be taken to be bipartite $[1, 2]$-factors.

Theorem 5 gives a similar weakening of Conjecture A, implying that every cubic graph has a decompostion into a spanning tree and a $[1, 2]$-factor of a subgraph. With Theorem 6 we weaken one factor and strengthen the other, showing that every cubic graph has a decomposition into a forest and a $[1, 2]$-factor. In Theorem 7 we observe that this statement is not likely an immediate corollary of the conjecture by showing that Conjecture A, if it holds, holds for subcubic graphs also, while Theorem 6 does not. Lemma 9 strengthens the result we get from Theorem 6 by showing that every cubic graph has a decomposition into a spanning forest and a $\{1, 2\}$-factor. In these last two statements, the forest is necessarily a *path factor*: a spanning union of disjoint non-trivial paths. This leads us to Theorems 11 and 13 which both imply that any cubic graph has a decomposition into two path factors.

2. Preliminary Observations

Let Δ be the maximal degree of G.

Observation 1. *G contains a set of $\lceil(\Delta+1)/2\rceil$ matchings the union of which is A, where $G - A$ is bipartite.*

Proof. Partition V greedily into two parts X, Y by successively transferring vertices of degree at least $\Delta/2$ in $G[X]$ to Y, and vice versa. This process terminates as each transference increases the number of edges between X and Y; when it terminates, $G[X]$ and

$G[Y]$ both have maximal degree at most $\lceil(\Delta-1)/2\rceil$. Applying Vizing's theorem (see [1]) each of $G[X]$ and $G[Y]$ can be properly edge colored with at most $\lceil(\Delta+1)/2\rceil$ colors. Using the same colors on each side, we get a decomposition of $G[X]$ and $G[Y]$ into at most $\lceil(\Delta+1)/2\rceil$ matchings. □ □

Observation 2. *If $\Delta = 4$, and $K_5 \not\subseteq G$, then G contains 2 matchings with union A, such that $G - A$ is bipartite.*

Proof. We apply Brooks' theorem (e.g. see [1]) to obtain a 4-coloring of G. Then we partition V into unions X, Y of pairs of color classes, and proceed as in the proof above. □ □

Observation 3. *If $\Delta \leq 3$, then G contains a matching M for which $G - M$ is bipartite.*

Proof. In the argument for Observation 1 above, both $G[X]$ and $G[Y]$ have maximal degree at most 1 in this case, hence the union of their edge sets is a matching M. □ □

Observation 4. *If $\Delta = 4$, and G contains a 1-factor, then G contains a $[1,2]$-factor H such that $G - E(H)$ is bipartite.*

Proof. Let F be a 1-factor. Apply Observation 3 to $G - E(F)$ to obtain a matching M for which $G - (E(F) \cup M)$ is bipartite. □ □

Observation 5. *If G is cubic, then G is a union of two edge-disjoint bipartite $[1,2]$-factors.*

Proof. This follows from Vizing's theorem (see [1]) by taking pairs of color classes in an 4-edge-coloring of G. □ □

Since any tree contains vertices of degree at most one, it is clear that in the worst case at least $\Delta - 1$ matchings need to be removed to obtain a forest. This lower bound is sharp.

Theorem 6. *If G is connected, then G contains a set of $\Delta - 1$ matchings with union A, such that $G - A$ is a spanning tree.*

Proof. We may assume $\Delta \geq 3$. Let T be a spanning tree having the least number of pendant edges. Assume that e is any edge of $G - E(T)$, and let C be the unique cycle of $T + e$. Then C contains at least one vertex v of degree at least 3 in G. Let u be a vertex of C adjacent to v. Then the spanning tree $T + e - uv$ has at most one more pendant edge than $T + e$. By the choice of T, there are not more pendant edges in T than in $T + e - uv$. It follows that T has at most one more pendant edge than $T + e$, hence the ends of e do not both have degree 1 in T.

Consider the graph $G - E(T)$. Its maximal degree is at most $\Delta - 1$, and the vertices of degree equal to $\Delta - 1$ form an independent set, since they have degree 1 in T. By a classical theorem of Fournier [2, 3], since the set of vertices of maximal degree induces a forest, there exists an edge-coloring of $G - E(T)$ using $\Delta - 1$ colors. It follows that the edges of $G - E(T)$ can be partitioned into $\Delta - 1$ matchings. □ □

If every vertex has degree 2 or 3, then we can decompose G into a forest and a $[1,2]$-factor as follows. The latter may however contain cycles of odd length and is not necessarily a union of two matchings.

Theorem 7. *If G is a $[2,3]$-graph, then G contains a $[1,2]$-factor H such that $G - E(H)$ is a forest.*

Proof. The theorem is clearly true if G has no vertex of degree 3; we proceed by induction on the number of edges of G.

Let e be an edge of G.

If both ends of e have degree 3 in G, then $G - e$ is a $[2,3]$-graph, so by induction it has a $[1,2]$-factor H such that $F = G - e - E(H)$ is a forest. If either end of e has degree 2 in H, then e is a pendant edge in $F + e$, so $F + e$ is a forest. But $G - E(H) = F + e$, so H is the desired $[1,2]$-factor of G. We may now assume that both ends of e have degree 1 in H. In this case $H + e$ is a $[1,2]$-factor of G, and we are done.

If both ends of e have degree 2 in G, we let P be the maximal path in G that contains e and does not contain any 3-vertex. If both ends of P are incident to a common 3-vertex v, then consider a path that contains both P and v but no other 3-vertex of G, and which is maximal with these properties, and now let P be this path instead. In either case $G - V(P)$ is a $[2,3]$-graph, and by hypothesis it contains a $[1,2]$-factor H such that $F = G - V(P) - E(H)$ is a forest. Now $H \cup P$ is a $[1,2]$-factor of G, and $G - E(H \cup P)$ is the forest obtained by adding to F the two edges not in P but incident to its endpoints. These are either pendant to F, form a P_2 pendant to F in the case that v was an endpoint of P, or one of them is pendant to F and the other, between v and one endpoint of P, is disjoint from F.

In the remaining case, we assume that G is bipartite with every edge joining a vertex of degree 2 to a vertex of degree 3.

By a simple argument invoking Hall's theorem (e.g. see [1]), G contains a matching M of all its 3-vertices. Since $G - M$ is a $[1,2]$-factor, and $G[M]$ is a forest, this finishes the proof. □ □

We note that Theorem 7 does not extend to $[1,3]$-graphs, since $K_{1,3}$ would be a counterexample, among others.

Theorem 8. *If Conjecture A is true, then its conclusion holds for all connected graphs with maximal degree at most 3.*

Proof. Let G be a connected graph of maximal degree ≤ 3.

Let H be any $\{2,3\}$-graph with a single 2-vertex w (e.g. H is a K_4 with one edge subdivided). We add several pairwise disjoint copies of H to G to obtain a cubic graph G' as follows. We join each 2-vertex v of G to a copy of H by adding a new edge joining the copy of w to v. For each 1-vertex u of G we identify u with w in a copy of H.

By assumption, the resulting cubic graph G' has a spanning tree T' such that $G' - E(T')$ consists of disjoint vertices, edges, and cycles. It is not hard to see that for the spanning tree T of G with edge set $E(G) \cap E(T')$, $G - E(T)$ consists of disjoint vertices, edges, and cycles, as needed. □ □

3. Partitioning into a forest and regular subgraphs

First we show that Conjecture A is true in the Hamiltonian case.

Theorem 9. *Conjecture A is true for Hamiltonian cubic graphs.*

Proof. Assume that G is a cubic graph with a Hamilton cycle H. Among all cycles in G that contain exactly one chord of H as an edge, choose C without chords in G; say, choose C to be a shortest such cycle. Let A be the set of chords of H that have exactly one end on C. Then $T = G[(E(H) \setminus E(C)) \cup A]$ is a spanning tree with the desired property; indeed, $G - E(T)$ consists of the cycle C and independent chords of H that are disjoint from C. □ □

Let $\mathcal{G}_1$ be the set of $[1,3]$-graphs with at most one 1-vertex, and $\mathcal{G}_2$ be the set of $[2,3]$-graphs with at most two 2-vertices. A $\{1,2\}$-factor F of G is called a *good factor of* G if $G - E(F)$ is a forest, and every 2-vertex of G that is not a cut vertex has degree 2 in F.

Lemma 10. *If G belongs to $\mathcal{G}_1 \cup \mathcal{G}_2$, then G has a good factor.*

Proof. The proof is by induction on $|V|$.

Assume $G \in \mathcal{G}_2$.

First, we assume that G has a bridge $e = uv$ such that each component of $G - e$ contains at most one 2-vertex of G. If $d(u) = d(v) = 2$, then $G - u - v$ has exactly two vertices of degree 2 and belongs to $\mathcal{G}_2$, and we apply induction to obtain a good factor F' of $G - u - v$. We then let $F = F' \cup G[e]$. Otherwise one or both of $d(u), d(v)$ is equal to 3, and each component of $G - e$ belongs to $\mathcal{G}_1 \cup \mathcal{G}_2$. We apply induction to get good factors F' and F'' of the components, and we let $F = F' \cup F''$. It is easy to check in either case that F is a good factor of G.

We may now assume that $G \in \mathcal{G}_2$ has a block B containing the 2-vertices of G, call them u and v, and at least one additional vertex. Since B is a block, the minimal degree of B is equal to 2. Furthermore, every vertex of degree 2 in B and distinct from u and v has degree 3 in G and is incident with a bridge of G.

If B is a cycle, we apply induction to find $\{1,2\}$-factors for the components of $G - V(B)$, then add B as an additional cycle to their union to construct F.

Otherwise B is a subdivision of a cubic graph H. Let e and f be edges of H such that u and v belong to the paths of B that correspond to e and f. Petersen [8] proved that every cubic graph with at most two bridges has a perfect matching, hence also a 2-factor. It is observed, in for example [5], that this implies that for any two edges of a 2-edge-connected

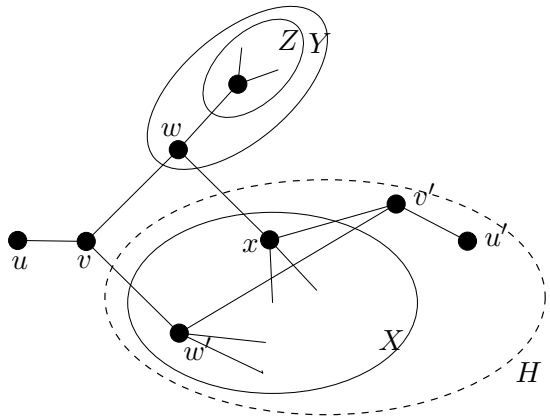

FIGURE 1. Sets X, Y and Z and graph H in proof of Lemma 9

cubic graph there exists a 2-factor containing them both. It follows that H has a 2-factor F_H that contains e and f. Let F_B be the 2-regular subgraph of B which corresponds to F_H after subdividing H into B. In particular F_B contains both u and v. We apply induction to the components of the subgraph $G - V(F_B) - E(B)$. Each 2-vertex of B that is not included in F_B is contained in one of these components. We let F be the union of the $\{1, 2\}$-factors of the individual components together with F_B. It is easy to check that F is a good factor of G.

Now assume $G \in \mathcal{G}_1 \setminus \mathcal{G}_2$.

Then G has exactly one 1-vertex, and all other vertices are 3-vertices. Let u be the 1-vertex of G, let v be its neighbor, and let w and w' be the neighbors of v in $G - u$.

If w is not a cut vertex of $G - u - v$, then by hypothesis, there exists a good factor F' of $G - u - v$. As F' is a good factor, the 2-vertex w of $G - u - v$ has degree 2 in F'. Clearly $F = F' \cup G[uv]$ is a $\{1, 2\}$-factor of G. Moreover, $G - E(F)$ is obtained from the forest $G - u - v - E(F')$ by adding edges vw and vw', where w has degree 1 in $G - E(F)$ and v has degree 1 in $G - E(F) - w$, and thus neither vertex belongs to a cycle of $G - E(F)$, so we deduce that F is good.

We can now assume that w is a cut vertex of $G - u - v$. Let Z be a component of $G - u - v - w$ that does not contain w', let $X = G - u - v - w - Z$, and let $Y = G - u - v - X = G[V(Z) \cup \{w\}]$. We observe $Y \in \mathcal{G}_1$ with w the 1-vertex of Y. (See Figure 1.) Let x be the neighbor of w in X. If $x = w'$ then we apply induction to $G - u - v - w - x \in \mathcal{G}_2$, and we obtain a good factor of G containing uv and wx. So we can assume $x \neq w'$.

Let H be the graph obtained from X by adding two new vertices u' and v' and edges $u'v'$, $v'w'$ and $v'x$. Then we have $H \in \mathcal{G}_1$, and H is of smaller order than G. We apply induction to H and to Y to find good factors F' and F'' of H and Y respectively. Let $F = (F' - u' - v') \cup F'' \cup G[uv]$. Then F is clearly again a $\{1,2\}$-factor of G. As $H - E(F')$ is a forest, the edges $v'w'$ and $v'x$ do not belong to cycles of $H - E(F')$. Moreover, they are themselves in $H - E(F')$ as $u'v'$ must be in F'. Hence x and w' are not connected in $(H - E(F')) \cap G$. Thus vw', vw and wx do not belong to cycles of $G - E(F)$, nor does any edge of $X - E(F')$ or $Y - E(F'')$. Thus F is a good factor of G. This concludes the proof by induction. □ □

Theorem 11. *Assume that G is cubic and let e be any edge of G. Then there exist $\{1,2\}$-factors F and F' of G with $e \in E(F) \setminus E(F')$ for which $G - E(F)$ and $G - E(F')$ are acyclic $[1,2]$-factors.*

Proof. Let u and v be the ends of e in G. Let H be the graph obtained from $G - e$ by adding new vertices x, y and edges ux, xy, and yv. Clearly H is in $\mathcal{G}_2$. If e is not a bridge of G, then by Lemma 10 there exists a good factor of H in which x and y have degree 2. Replacing the path $uxyv$ by e we obtain the good factor F of G with $e \in F$. In the case when e is a bridge of G, we consider $H - xy$ instead. The components of $H - xy$ belong to $\mathcal{G}_1$, and using Lemma 10 they have good factors, which contain ux and yv. We obtain F from their union by replacing ux and yv by uv.

Now let H' be the graph obtained from $G - e$ by adding new vertices x, y and edges ux, vx, and xy. Then H' is in $\mathcal{G}_1$. By Lemma 10 there exists a $\{1,2\}$-factor of H', from which we obtain F' by deleting x and y. □ □

Since every component of a $\{1,2\}$-graph is a K_2 or a cycle, the following is immediate from Theorem 11.

Corollary 12. *Every cubic graph G contains a spanning forest F for which every component of $G - E(F)$ is a K_2 or a cycle.*

By a similar argument as in the proof of Theorem 8 we then obtain:

Corollary 13. *Every nontrivial connected graph G with maximal degree at most 3 contains a spanning forest F such that every component of $G - E(F)$ is a K_1, a K_2 or a cycle.*

4. Path-factorizations

A *path factor* in G is a $[1,2]$-factor of G without cycles, that is, a spanning subgraph in which every component is a nontrivial path. For $k \in \mathbb{N}$, a *k-path factorization* of G is a set of k path factors the edge sets of which form a partition of E.

We will apply a theorem of Gupta [4]. It states that if a graph has minimal degree $\delta > 1$, then there exists an improper coloring of its edges using $\delta - 1$ colors, such that every color appears at each vertex.

Theorem 14. *Assume that G is r-regular for $r \geq 3$. Then G has an $(r-1)$-path factorization.*

Proof. Applying Gupta's theorem with $\delta = r$ to G, there exists a partition of E into $r-1$ disjoint sets, such that each vertex is incident with at least one edge from every set. Since each vertex has degree r, it is clear that every set induces a $[1,2]$-factor. So the edges of G partition into $r-1$ $[1,2]$-factors. Among all such partitions we choose one in which the total number of cycles in all its factors is minimal. Suppose that one of the $[1,2]$-factors F contains a cycle C, and let F' be any one of the $r-2>0$ remaining $[1,2]$-factors. Then every vertex of C has degree 2 in F and degree 1 in F', in particular such a vertex is an endvertex of a path component of F'. Since C has length at least 3, there is an edge $e = uv$ of C such that u and v are endvertices of distinct paths in F'. But then we replace F and F' in the factorization into $[1,2]$-factors by $F-e$ and $F'+e$, and we obtain a factorization with fewer cycles, a contradiction. We deduce that none of the $r-1$ $[1,2]$-factors has cycles, hence they define a path partition. □ □

Lemma 15. *If G is the union of a cycle C and a (possibly trivial) path Q with endvertices u and v, such that C and Q are edge-disjoint and $V(C) \cap V(Q) = \{u\}$, then G is the edge-disjoint union of a path factor of G with a path factor of $G-v$.*

Proof. Let $e = uw$ be an edge of C incident to u. Then $G-e$ is a path Q' with ends v and w. From the color classes of the proper edge 2-coloring of Q', we obtain the edge sets of a path factor of G and of a path factor of $G-v$ by adding e to the color class that does not contain an edge incident to w. □ □

Theorem 16. *If G is a $[2,3]$-graph, then G has a 2-path factorization if and only if no component of G is an odd cycle.*

Proof. It is clear that if a 2-path factorization exists, and if v is a vertex of degree 2, then the two edges incident to v belong to different factors, and the factorization of G is a 2-path factorization. In particular any 2-path factorization of a 2-regular graph corresponds to a proper edge 2-coloring. It follows that a graph with an odd cycle component has no path factorizations.

Assume that G is a $[2,3]$-graph without an odd cycle component. We prove by induction on $|E|$ that G has a 2-path factorization.

We may assume that G is connected, otherwise we apply induction to the components of G. We can also assume that G is not 2-regular, since otherwise G is an even cycle, hence G has a 1-factorization and we are done.

By Theorem 11 we may assume that G is not cubic. Let P be a maximal path in G all vertices of which have degree 2 in G. Then P contains a 2-vertex of G, and $G' = G - V(P)$ has fewer edges than G.

If G' is a $[2,3]$-graph without odd cycle component, by hypothesis G' has a path factorization into factors F_1 and F_2. We edge 2-color the path induced by the edges $E(G)\setminus E(G')$. Adding one color class to F_1, the other to F_2, we get a path factorization of G.

If G' has an odd cycle component C, then $H = G[V(C) \cup V(P)] - vz$ satisfies the assumptions of Lemma 15, with v an endvertex of P and z a vertex of G adjacent to v. Hence there exist a path factor F_1 of H and a path factor F_2 of $H - v$, such that $E(F_1)$ and $E(F_2)$ are disjoint and have union $E(H)$. If $G - V(H)$ is not an odd cycle, then by the induction hypothesis there exists a 2-path factorization of $G - V(H)$ and we extend F_1 and $F_2 + vz$ to path factors of G, to obtain a 2-path factorization of G. If $G - V(H)$ is an odd cycle C', then we apply Lemma 15 to $C' + zv$ instead, and we combine the path factor on either side which does not contain v with the path factor on the other side which contains v, to get a path factorization of G.

If G' is not a $[2,3]$-graph, then G has a 3-vertex v adjacent to both ends of P. Let Q be the unique path in G which has v as one end, a 3-vertex u of G distinct from v as the other end, and in which all other vertices have degree 2. Applying Lemma 15 we get edge disjoint path factors for $G[V(P) \cup V(Q)]$ and $G[V(P) \cup V(Q)] - u$. There are edge disjoint path factors either of the remaining subgraph $G - (V(P) \cup V(Q) \setminus \{u\})$, or one of it, and one of $G - (V(P) \cup V(Q))$, which we may combine these with to obtain the desired path factors of G. Indeed, $G - (V(P) \cup V(Q) \setminus \{u\})$ is clearly a connected $[2,3]$-graph; if it is not an odd cycle then it has a 2-path factorization, by induction. If it is an odd cycle, we obtain the desired path factors from Lemma 15. □ □

As no graph with a leaf can have a 2-path factorization, this is equivalent to the following.

Corollary 17. *A graph of maximal degree at most* 3 *has a* 2*-path factorization if and only it contains no leaves and no components that are odd cycles.*

5. Concluding Remarks

Our results are related to, and mostly motivated by Conjectures A and B. In particular, in Theorem 9 we observe that Conjecture A holds in the case that the graph G is Hamiltonian. It would be nice to see this proof extended to the case that G contains a Hamilton path.

It has recently come to our attention that Conjecture A has been proved in the case that G is a planar graph, see [7]. We observe that Conjecture B is also known for planar graphs as it is a consequence of the Four Color Theorem that every bridgeless planar cubic graph has a 3-edge-coloring.

The observations of Section 2 suggest the following question.

Question 18. *How many matchings must one remove from a graph of maximal degree* Δ *to leave a bipartite graph?*

It is shown in Section 2 that if Δ is even then $\Delta/2+1$ is enough, and that if Δ is odd then $(\Delta+1)/2$ is enough.

This is tight in several cases, as one can see by considering the clique $K = K_{\Delta+1}$. In any bipartition $A \cup B$ of the vertices of K defined by the bipartite graph H, all edges in the cliques K_A and K_B induced by A and B must be in some matching. It is known that K_A can be decomposed into $|A|-1$ matchings if $|A|$ is even, and requires $|A|$ matchings if $|A|$ is odd. If Δ is 0 modulo 4, then $\max(|A|,|B|)$ is minimised with $|B| = |A|+1 = \Delta/2+1$, which is odd, so this number of matchings is required.

Similar analysis for the other values of Δ modulo 4 show that our upper bounds on the number of matchings needed are tight when Δ is congruent to 0 or 1 modulo 4, but may exceed the actual value by one when Δ is congruent to 2 or 3 modulo 4.

In particular, we ask the following.

Question 19. *From any graph of maximal degree* 6, *can one remove three matchings and leave a bipartite graph?*

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